Notes on covariance penalties

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1 Introduction

In estmation theory there is a great importance in evaluating the prediction error of the model. In the majority of cases cross-validation suffices. However, when the dataset is "small" cross-validation estimates this error poorly, in this case the partition of the validation set becomes a deciding factor of the error estimate. A way to combat this is to look at covariance penalties which offers a more deterministic approach to the errro estimate with the cost of having additional assumptions.

If we focus specifically on trees we can show some interesting? findings...

2 Independent copy construction

Consider iid trainingsdata $((X_i, Y_i))_{i=1}^n$ and independent test response data Y^*. Starting of with the general expression for prediction error

$$\mathbf{MSEP} = \mathbb{E}\left((Y^* - \widehat{\mu})^2\right), \tag{1}$$

where the $*$ denots the data being an independent copy of the trainingsdata used to create the estimator $\widehat{\mu}$. Using this general expression for prediction error we can manipulate it in the following way

$$\begin{aligned}\mathbb{E}\left((Y^* - \widehat{\mu})^2\right) &= \mathbb{E}\left((Y^* - \mu + \mu - \widehat{\mu})^2\right) &(2)\\ &= \mathbb{E}\left((Y^* - \mu)^2\right) + 2\mathbb{E}\left((Y^* - \mu)(\widehat{\mu} - \mu)\right) + \mathbb{E}\left((\widehat{\mu} - \mu)^2\right), &(3)\end{aligned}$$

where $\mu(x) = \mathbb{E}[Y \mid X = x]$ is the true regression function. Since we assume that Y^* and $\widehat{\mu}$ are independent the second term in the above expansion is identically zero. Furthermore, since $\widehat{\mu} = \frac{1}{n}\sum_{j=1}^n Y_j$, we can express the prediction error as

$$\begin{aligned}\mathbf{MSEP} &= \mathbb{E}\left((Y^* - \mu)^2\right) + 2\mathbb{E}\left((Y^* - \mu)(\widehat{\mu} - \mu)\right) + \mathbb{E}\left((\widehat{\mu} - \mu)^2\right) &(4)\\ &= \mathbb{E}\left((Y^* - \mu)^2\right) + \mathbb{E}\left((\frac{1}{n}\sum_{j=1}^n Y_j - \mu)^2\right) &(5)\\ &= \sigma^2 + \frac{\sigma^2}{n}, &(6)\end{aligned}$$

where $\sigma^2 := \mathbb{E}(\sigma^2(X))$ (in the heteroscedastic case). One can interepret this result as the prediction error on a specific leaf of a tree. In the most simple case one can see this as the prediction error when there is no split.

Using simple probability one can once again manipulate expression (**??**).

$$\mathbf{MSEP}_{split} = \mathbb{E}\left((Y^* - \widehat{\mu})^2\right) \tag{7}$$
$$= \mathbb{E}\left((Y^* - \widehat{\mu})^2 \middle| X \in \mathcal{A})\mathbb{P}(X \in \mathcal{A}\right) \tag{8}$$
$$+ \mathbb{E}\left((Y^* - \widehat{\mu})^2 \middle| X \in \mathcal{A}^c)\mathbb{P}(X \in \mathcal{A}^c\right). \tag{9}$$

Performing similar analysis as done in (**??**), we get the expression

$$\mathbf{MSEP}_{split} = \sigma^2_{\mathcal{A}}(1 + \frac{1}{n_{\mathcal{A}}})\pi_{\mathcal{A}} + \sigma^2_{\mathcal{A}^c}(1 + \frac{1}{n_{\mathcal{A}^c}})\pi_{\mathcal{A}^c}, \tag{10}$$

where

$$\sigma^2_{\cdot} := \mathrm{Var}\left(Y \mid X \in \cdot\right),$$

$\pi_{\cdot}, n_{\cdot}$ is the variance, probability and number of samples respectively. An interpretation of this is having a binary split of the data.

A natural question when building tree models is, "should a split be made on this node?". One way to make a decision about this is to compare if the prediction error when making the split is lower than the non-split error, that is

$$\mathbf{MSEP}_{split} > \mathbf{MSEP}. \tag{11}$$

If the inequality holds true then a split should **not** be made, since doing so will increase the prediction error. The rule in (**??**) can be written as

$$\sigma^2_{\mathcal{A}}(1 + \frac{1}{n_{\mathcal{A}}})\pi_{\mathcal{A}} + \sigma^2_{\mathcal{A}^c}(1 + \frac{1}{n_{\mathcal{A}^c}})\pi_{\mathcal{A}^c} > \sigma^2(1 + \frac{1}{n}). \tag{12}$$

Note that $n = n_{\mathcal{A}} + n_{\mathcal{A}^c}$.

3 Insample construction

In this case we will look at a construction which is based on the insample error. We assume homoscedacity, i.e. $\mathrm{Var}(Y \mid X) = \sigma^2$ is non-random. Fix $x_1, \ldots, x_N$ and let $\mathcal{L}(Y_i) = \mathcal{L}(Y_i^*) = \mathcal{L}\left(Y \mid X = x_i\right)$, with indpendence between Y_i^* and Y_i, $i = 1, \ldots, n$. Let $\hat{\mu}(x) = \hat{\mu}(x, (x_i, Y_i)_{i=1}^N)$ be an estimate of μ depending on the trainingsdata. Following Olas lecture notes,

$$\mathbf{MSEP}_{\mathrm{in}} = \frac{1}{N}\sum_{i=1}^{N}(Y_i^* - \hat{\mu}(x_i))^2, \quad \mathbf{MSE} = \frac{1}{N}\sum_{i=1}^{N}(Y_i - \hat{\mu}(x_i))^2,$$

and we get that

$$
\begin{aligned}
&\mathbb{E}[\mathbf{MSEP}_{\text{in}}] - \mathbb{E}[\mathbf{MSE}] \\
=&\frac{1}{N}\sum_{i=1}^{N}\mathbb{E}\left[(Y_i^* - \hat{\mu}(x_i))^2 - (Y_i - \hat{\mu}(x_i))^2\right] \\
=&\frac{1}{N}\sum_{i=1}^{N}\left(\mathbb{E}[(Y_i^*)^2] - \mathbb{E}[(Y_i)^2] + \mathbb{E}[\hat{\mu}(x_i)^2] - \mathbb{E}[\hat{\mu}(x_i)^2] + 2\left(\mathbb{E}\left[Y_i\hat{\mu}(x_i)\right] - \mathbb{E}\left[Y_i^*\hat{\mu}(x_i)\right]\right)\right) \\
=&\frac{2}{N}\sum_{i=1}^{N}\left(\mathbb{E}\left[Y_i\hat{\mu}(x_i)\right] - \mathbb{E}\left[Y_i\right]\mathbb{E}\left[\hat{\mu}(x_i)\right]\right) = \frac{2}{N}\sum_{i=1}^{N}\operatorname{Cov}(Y_i, \hat{\mu}(x_i)).
\end{aligned}
$$

In the second-last equality, we have used the independence between Y_i^* and $\hat{\mu}(x_i)$ and the fact that $\mathbb{E}[Y_i^*] = \mathbb{E}[Y_i]$. Hence,

$$\mathbb{E}[\mathbf{MSEP}_{\text{in}}] = \mathbb{E}[\mathbf{MSE}] + \frac{2}{N}\sum_{i=1}^{N}\operatorname{Cov}(Y_i, \hat{\mu}(x_i)).$$

If the estimated regression function is a balanced tree, then $\hat{\mu}(x) = \sum_{j=1}^{J} I\{x \in A_j\}\hat{m}_j$, where $A_1, \ldots, A_J$ form a partition of the covariate space and $\hat{m}_j = \frac{1}{|A_j|}\sum_{x_i \in A_j} Y_i$ with $|A_j| := |\{i \colon x_i \in A_j\}|$. Writing $\hat{\mu}(x_i) = \hat{\mu}_i$, the summands above become

$$
\begin{aligned}
\operatorname{Cov}\left(Y_i, \widehat{\mu}_i\right) &= \operatorname{Cov}\left(Y_i, \frac{1}{|A_{j(i)}|}\sum_{k \in A_{j(i)}} Y_k\right) && (13) \\
&= \frac{1}{|A_{j(i)}|}\operatorname{Var}\left(Y_i\right) = \frac{1}{|A_{j(i)}|}\sigma^2. && (14)
\end{aligned}
$$

Here, $j(i)$ is the index of the region containing x_i. Altogether, this gives

$$\mathbb{E}[\mathbf{MSEP}_{\text{in}}] = \mathbb{E}[\mathbf{MSE}] + \frac{2\sigma^2 J}{N},$$

since $\sum_{i=1}^{N}\frac{1}{|A_{j(i)}|} = J$ is the number of disjoint regions.

The comparison criterion in this case now simply becomes, in the binary split case

$$\sigma_{\mathcal{A}}^2(n_{\mathcal{A}} + 2) + \sigma_{\mathcal{A}^c}^2(n_{\mathcal{A}^c} + 2) > \sigma^2(n + 2). \tag{15}$$

Note that this is an equivalent representation of the splitting criterion as one found in the previous section. This can be clear by simplifying the MSE term. To begin with, we can set $J = 1$, assuming here that we don't want to make a split. Now,

$$\begin{aligned}
\mathbb{E}[\mathbf{MSEP}_{\text{in}}] &= \mathbb{E}[\mathbf{MSE}] + \frac{2\sigma^2}{N} \\
&= \frac{1}{N}\sum_{1}^{N}(Y_i - \widehat{\mu}_i)^2 + \frac{2\sigma^2}{N} \\
&= \frac{1}{N}\sum_{1}^{N}(Y_i - \mu_i + \mu_i - \widehat{\mu}_i)^2 + \frac{2\sigma^2}{N}
\end{aligned}$$

where, μ_i is the true underlying mean. We can now expand the square term

$$\begin{aligned}
&\frac{1}{N}\mathbb{E}\left(\sum_{1}^{N}(Y_i - \mu_i) - 2(Y_i - \mu_i)(\widehat{\mu}_i - \mu_i) + (\widehat{\mu}_i - \mu_i)^2\right) + \frac{2\sigma^2}{N} \\
&= \sigma^2 - \frac{2}{N}\sum_{1}^{N}\operatorname{Cov}(Y_i, \widehat{\mu}_i) + \frac{\sigma^2}{N} + \frac{2\sigma^2}{N}.
\end{aligned}$$

Here, the covariance expression does not vanish since everythin is insample. Simplifying the covariance term we get

$$\begin{aligned}
&\sigma^2 - \frac{2}{N}\sum_{1}^{N}\operatorname{Cov}(Y_i, \widehat{\mu}_i) + \frac{\sigma^2}{N} + \frac{2\sigma^2}{N} \\
&= \sigma^2 - \frac{2}{N}\sigma^2 + \frac{\sigma^2}{N} + \frac{2\sigma^2}{N} = \frac{\sigma^2(N+1)}{N},
\end{aligned}$$

which is identical to the expression found in the previous section.

4 Distribution of stopping variable

Consider $n = n_A + n_B$ datapoints $(x_i, Y_i)_{i=1}^{n}$, $\mathcal{L}(Y_i) = \mathcal{L}(Y \mid X = x_i) = \text{Normal}(\mu(x_i), \sigma^2)$. In other words, writing $\mu_i = \mu(x_i)$, (Y_i) follows a multivariate normal distribution with mean

$$\mu = (\mu_1, \ldots, \mu_n)^T, \quad \Sigma = \sigma^2 I_n.$$

Consider two disjoint sets A, B with $x_1, \ldots, x_{n_A} \in A$ and $x_{n_A+1}, \ldots, x_n \in B$ and let $C := A \dot{\cup} B$. Further, set

$$\sigma_A^2 = \operatorname{Var}(Y \mid X \in A) = \mathbb{E}[(Y - \mathbb{E}[Y \mid X \in A])^2 \mid X \in A]$$

Note that in general $\sigma^2 \leq \sigma_A^2$. Define σ_B^2 and σ_C^2 likewise. Estimating π_A by n_A/n and σ_A^2 by the unbiased variance estimator

$$\hat{\sigma}_A^2 = \frac{1}{n_A - 1}\sum_{i=1}^{n_A}(Y_i - \bar{Y}_A)^2,$$

where $\bar{Y}_A = 1/n_A \sum_{i=1}^{n_A} Y_i$, and analogously for the sets B, C, and plugging this into the above motivated (**??**), we arrive at the stopping criterion

$$Z := \tag{16}$$

$$\frac{n_A + 1}{n_A - 1} \sum_{i=1}^{n_A} (Y_i - \bar{Y}_A)^2 + \frac{n_B + 1}{n_B - 1} \sum_{i=n_A+1}^{n} (Y_i - \bar{Y}_B)^2 - \frac{n+1}{n-1} \sum_{i=1}^{n} (Y_i - \bar{Y})^2 > 0. \tag{17}$$

Let

$$D_A = \left[\begin{array}{ccc|c} 1 & \cdots & 1 & \\ & \vdots & & \mathbf{0} \\ 1 & \cdots & 1 & \\ \hline & \mathbf{0} & & \mathbf{0} \end{array}\right], \quad I_A = \left[\begin{array}{c|c} \mathbf{I}_{n_A} & \mathbf{0} \\ \hline \mathbf{0} & \mathbf{0} \end{array}\right] \tag{18}$$

and D_B, I_B defined correspondingly, with the one-matrix and the identity matrix in the lower right block instead. We can then rewrite (**??**) as a quadratic form

$$Z = Y^T Q Y,$$

where

$$\begin{aligned} Q =& \frac{n_A + 1}{n_A - 1} \left(I_A - \frac{1}{n_A} D_A \right)^T \left(I_A - \frac{1}{n_A} D_A \right) + \\ & \frac{n_B + 1}{n_B - 1} \left(I_B - \frac{1}{n_B} D_B \right)^T \left(I_B - \frac{1}{n_B} D_B \right) - \\ & \frac{n+1}{n-1} \left(I_n - \frac{1}{n} \mathbf{1} \right)^T \left(I_n - \frac{1}{n} \mathbf{1} \right). \end{aligned}$$

Here, $\mathbf{1}$ is the matrix of ones only. Using that all matrices in brackets are symmetric and idempotent,

$$Q = \frac{n_A + 1}{n_A - 1} \left(I_A - \frac{1}{n_A} D_A \right) + \frac{n_B + 1}{n_B - 1} \left(I_B - \frac{1}{n_B} D_B \right) - \frac{n+1}{n-1} \left(I_n - \frac{1}{n} \mathbf{1} \right) =$$

$$\left[\begin{array}{ccc|ccc} \frac{n_A+1}{n_A} - \frac{n+1}{n} & \frac{n+1}{(n-1)n} - \frac{n_A+1}{(n_A-1)n_A} & \cdots & & & \\ \frac{n+1}{(n-1)n} - \frac{n_A+1}{(n_A-1)n_A} & & & \frac{n+1}{(n-1)n} & \cdots & \\ \vdots & \ddots & & \vdots & & \\ \hline & & & \frac{n_B+1}{n_B} - \frac{n+1}{n} & \frac{n+1}{(n-1)n} - \frac{n_B+1}{(n_B-1)n_B} & \cdots \\ \frac{n+1}{(n-1)n} & \cdots & & \frac{n+1}{(n-1)n} - \frac{n_B+1}{(n_B-1)n_B} & & \\ \vdots & & & \vdots & \ddots & \end{array}\right].$$

Q has four distinct eigenvalues

$$\begin{aligned}
\lambda_A =& \frac{n_A+1}{n_A-1} - \frac{n+1}{n-1} = \frac{2n_B}{(n_A-1)(n-1)} > 0,\\
\lambda_B =& \frac{n_B+1}{n_B-1} - \frac{n+1}{n-1} = \frac{2n_A}{(n_B-1)(n-1)} > 0,\\
\lambda_{\text{neg}} =& -\frac{n+1}{n-1} < 0,\\
\lambda_0 =& 0
\end{aligned}$$

with corresponding eigenspaces

$$\begin{aligned}
\mathrm{Eig}_{\lambda_A} =& \mathrm{span}\left(\begin{bmatrix} 1\\ -1\\ 0\\ \vdots\\ 0\\ \mathbf{0}_{n_B} \end{bmatrix}, \ldots, \begin{bmatrix} 1\\ 0\\ \vdots\\ 0\\ -1\\ \mathbf{0}_{n_B} \end{bmatrix} \right), \quad \mathrm{Eig}_{\lambda_B} = \mathrm{span}\left(\begin{bmatrix} \mathbf{0}_{n_A}\\ 1\\ -1\\ 0\\ \vdots\\ 0 \end{bmatrix}, \ldots, \begin{bmatrix} \mathbf{0}_{n_A}\\ 1\\ 0\\ \vdots\\ 0\\ -1 \end{bmatrix} \right),\\
\mathrm{Eig}_{\lambda_{\text{neg}}} =& \mathrm{span}\left(\begin{bmatrix} n_B\\ \vdots\\ n_B\\ -n_A\\ \vdots\\ -n_A \end{bmatrix} \right), \quad \mathrm{Eig}_{\lambda_0} = \mathrm{span}\left(\begin{bmatrix} 1\\ \vdots\\ 1 \end{bmatrix} \right)
\end{aligned} \tag{19}$$

and hence geometric multiplicities $n_A-1, n_B-1, 1$ and 1, respectively. Let

$$P = \left[\begin{array}{c|c|c|c}
P_{11} & \mathbf{0} & \begin{matrix} n_B/\sqrt{n_A^2 n_B + n_A n_B^2}\\ \vdots\\ n_B/\sqrt{n_A^2 n_B + n_A n_B^2} \end{matrix} & \begin{matrix} 1/\sqrt{n}\\ \vdots\\ 1/\sqrt{n} \end{matrix}\\
\hline
\mathbf{0} & P_{22} & \begin{matrix} -n_A/\sqrt{n_A^2 n_B + n_A n_B^2}\\ \vdots\\ -n_A/\sqrt{n_A^2 n_B + n_A n_B^2} \end{matrix} & \begin{matrix} 1/\sqrt{n}\\ \vdots\\ 1/\sqrt{n} \end{matrix}
\end{array}\right],$$

where P_{11} (P_{22}) is an $n_A \times n_A - 1$ ($n_B \times n_B - 1$) matrix obtained by applying the Gram-Schmidt orthogonalization to the first n_A (last n_B) components of the vectors spanning Eig_{λ_A} (Eig_{λ_B}) from (**??**) above.

By Mathai, Provost, "Quadratic forms in random variables", page 29, formula (3.1a.4), it follows that

$$\begin{aligned}
Z =& Y^T Q Y\\
\overset{d}{=}& \frac{\sigma^2}{n-1}\left(\frac{2n_B}{n_A-1}\sum_{i=1}^{n_A-1}(U_i+b_i)^2 + \frac{2n_A}{n_B-1}\sum_{i=n_A}^{n-2}(U_i+b_i)^2 - (n+1)(U_{n-1}+b_{n-1})^2 \right).
\end{aligned} \tag{20}$$

Here, $U_1, \ldots, U_{n-1}$ are iid standard normally distributed and, with $\mu^T = (\mu_A^T, \mu_B^T)$,

$$b = \frac{1}{\sigma} P^T \mu = \frac{1}{\sigma} \begin{bmatrix} P_{11}^T \mu_A \\ P_{22}^T \mu_B \\ \left(n_B \sum_{i=1}^{n_A} \mu_i - n_A \sum_{i=n_A+1}^{n} \mu_i\right) / \sqrt{n_A^2 n_B + n_A n_B^2} \\ \left(\sum_{i=1}^{n} \mu_i\right) / \sqrt{n} \end{bmatrix}.$$

Non-central Chi-square Let $X_1, \ldots, X_m$ independent normally distributed random variables with means $a_1, \ldots, a_m$ and unit variances and set $\xi = \sum_{i=1}^m a_i^2$ (non-centrality parameter). Then, $\sum_{i=1}^m X_i^2$ follows a non-central chi-squared distribution with m degrees of freedom and is denoted by $\chi_m^2(\xi)$.

In order to find the limit in (**??**) in terms of independent non-central chi-squares, we need to compute

$$\xi_A = \sum_{i=1}^{n_A - 1} b_i^2, \quad \xi_B = \sum_{i=n_A}^{n-2} b_i^2, \text{ and } \quad \xi_{\text{neg}} = b_{n-1}^2.$$

Denote by $\tilde{P}_{11}^T$ the orthogonal $n_A \times n_A$ matrix obtained by adding $(1/\sqrt{n_A}, \ldots, 1/\sqrt{n_A}) \in \mathbb{R}^{n_A}$ as an additional row to P_{11}^T. We then get

$$\begin{aligned} \xi_A = \sum_{i=1}^{n_A - 1} b_i^2 =& \frac{1}{\sigma^2} \left(\sum_{i=1}^{n_A} \left(\left(\tilde{P}_{11}^T \mu_A \right)_i \right)^2 - |(1/\sqrt{n_A}, \ldots, 1/\sqrt{n_A}) \mu_A|^2 \right) \\ =& \frac{1}{\sigma^2} \left(\sum_{i=1}^{n_A} \mu_i^2 - \frac{1}{n_A} \left(\sum_{i=1}^{n_A} \mu_i \right)^2 \right), \end{aligned}$$

where we have used the orthogonality of $\tilde{P}_{11}$ in the last equality. Similarly, one gets

$$\xi_B = \frac{1}{\sigma^2} \left(\sum_{i=n_A}^{n-2} \mu_i^2 - \frac{1}{n_B} \left(\sum_{i=n_A}^{n-2} \mu_i \right)^2 \right)$$

and

$$\xi_{\text{neg}} = \frac{\left(n_B \sum_{i=1}^{n_A} \mu_i - n_A \sum_{i=n_A+1}^{n} \mu_i\right)^2}{\sigma^2 \left(n_A^2 n_B + n_A n_B^2\right)}.$$

Altogether, we can rewrite (**??**) to get

$$Z \stackrel{d}{=} \frac{\sigma^2}{n-1} \left(\frac{2 n_B}{n_A - 1} \chi_{n_A - 1}^2(\xi_A) + \frac{2 n_A}{n_B - 1} \tilde{\chi}_{n_B - 1}^2(\xi_B) - (n+1) \bar{\chi}_1^2(\xi_{\text{neg}}) \right), \quad (21)$$

where $\chi_{n_A-1}^2(\xi_A), \tilde{\chi}_{n_B-1}^2(\xi_B)$ and $\bar{\chi}_1^2(\xi_{\text{neg}})$ are independent. In the special case where $\mu_A = (c, \ldots, c)^T$ is constant, we get that $\xi_A = 0$ (same for μ_B). If $\mu = (c, \ldots, c)^T$, then $\xi_{\text{neg}} = \xi_A = \xi_B = 0$, i.e. we get central chi squared distributions.

Hypothesis test In practice, one would check the criterion we have mentioned above and decide if a split is good or bad. That is, compare the msep after split to the msep without split. However, one should also account for randomness. Naturally, we should not accept the criterion based on the outcome of the criterion but instead, we would like to have some certainty or confidence in the outcome before it can be accepted. Say, we want to be $1-\alpha\%$ certain that the outcome of Z is not noise. This leads to hypothesis testing. We state the following null-hypothesis

$$H_0 : \mu_A = \mu_B = \mu.$$

The null-hypothesis states that there is no difference in the mean function in the two seperate regions. We then get that the distriubtion of Z under H_0 call it Z_0 and it is is distributed as

$$Z_0 \overset{d}{=} \frac{\sigma^2}{n-1}\left(\frac{2n_B}{n_A-1}\chi^2_{n_A-1} + \frac{2n_A}{n_B-1}\tilde{\chi}^2_{n_B-1} - (n+1)\bar{\chi}^2_1\right).$$

In fact, given the null-hyptohesis H_0, the individual Chi-Squared random variables are centered as well. If the null-hypothesis is true we would expect the outcome of Z to be close to 0. In contrast, if H_0 is untrue, then, we expect the outcome of Z to be very negative. If the out come of Z is realised in the p quantile of Z_0, we reject H_0 since the outcome is too extreme under H_0. Find analytical quantile!

In a greedy approach, one stops splitting as soon as the criterion (**??**) is met that is

$$Z > 0.$$

Or equivalently, continue splitting for as long as the criterion

$$Z \leq 0,$$

is satisfied. Using this simple criterion may be bad for generalization since we are not penalizing a model that is more complex. In practice, a tolerance can be chosen so that the criterion is more strict. That is, continue splitting as long as the criterion

$$Z \leq -\delta,$$

is satisfied. $\delta > 0$ is typically chosen heuristically as far as I know. This can be stated as splitting for as long as the msep decreases by atleast δ at each iteration. Larger delta is more restrictive. One can tweak δ to give better results. A dynamic *delta* can also be chosen.

Our approach is equivalent as we continue to split as long as the criterion

$$Z \leq q^0_\alpha$$

is satisfied. Here, q^0_α is the α quantile under the null-hyptohesis H_0. However, we chose the tolerance via hypothesis testing. Furthermore, the tolerance we choose will change based on the sample size in the splitting node and confidence level.

An issue with the approach mentioned, is that the parameter σ needs to be specified. An estimate can then be used. In the following section we will instead be looking at a ratio, which does suffer the same issue.

4.1 Ratio distribution

Let $H_0 : \mu = (c, \dots, c)^T$ for some $c \in \mathbb{R}$ and denote by $\mathbb{P}_0$ a probability measure satisfying H_0, i.e. $\mathbb{E}_0[Y] = (c, \dots, c)^T$ for some $c \in \mathbb{R}$. In order to find a statistic which does not depend on the unknown parameter σ^2, we would like to investigate the distribution of the ratio

$$R = \frac{\frac{n_A+1}{n_A-1}\sum_{i=1}^{n_A}(Y_i - \bar{Y}_A)^2 + \frac{n_B+1}{n_B-1}\sum_{i=n_A+1}^{n}(Y_i - \bar{Y}_B)^2}{\frac{n+1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2}$$

under $\mathbb{P}_0$. For this purpose, we note that for each $r > 0$,

$$\mathbb{P}_0(R > r) = \mathbb{P}_0(Z_r > 0),$$

where

$$Z_r = \frac{n_A+1}{n_A-1}\sum_{i=1}^{n_A}(Y_i - \bar{Y}_A)^2 + \frac{n_B+1}{n_B-1}\sum_{i=n_A+1}^{n}(Y_i - \bar{Y}_B)^2 - r\frac{n+1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2.$$

As above, we can write this as a quadratic form

$$Z_r = Y^T Q_r Y$$

and the eigenvalues of Q_r are given by

$$\begin{aligned}
\lambda_A^r =& \frac{n_A+1}{n_A-1} - r\frac{n+1}{n-1} = \frac{(n_A+1)(n-1) - r(n+1)(n_A-1)}{(n-1)(n_A-1)}, \\
\lambda_B^r =& \frac{n_B+1}{n_B-1} - r\frac{n+1}{n-1} = \frac{(n_B+1)(n-1) - r(n+1)(n_B-1)}{(n-1)(n_B-1)}, \\
\lambda_{\text{neg}}^r =& - r\frac{n+1}{n-1}, \\
\lambda_0^r =& 0
\end{aligned}$$

and the eigenspaces and geometric multiplicities are the same as for the corresponding Q-eigenvalues. Hence, under H_0,

$$Z_r \stackrel{d}{=} \sigma^2\left(\lambda_A^r \chi^2_{n_A-1} + \lambda_B^r \tilde{\chi}^2_{n_B-1} + \lambda_{\text{neg}}^r \bar{\chi}^2_1\right) \tag{22}$$

with independence between the central chi squares $\chi^2_{n_A-1}, \tilde{\chi}^2_{n_B-1}$ and $\bar{\chi}^2_1$. Denoting the random variable in brackets by $Z_{r,1}$, it follows that

$$\mathbb{P}_0(R > r) = \mathbb{P}_0(Z_r > 0) = \mathbb{P}_0(Z_{r,1} > 0).$$

4.2 Efficient computation of $F_R^0(r)$

Let like above

$$F_R^0(r) = \mathbb{P}_0(R \leq r) \tag{23}$$

denote the distribution function of the ratio R, given H_0 holds true. By the the last section together with [1971, Joseph L. Fleiss, On the Distribution of a Linear Combination of Independent Chi Squares, Corollary and following p.142], (**??**) is given by

$$\mathbb{P}_0(Z_{r,1} \leq 0) = \frac{\Gamma\left(\frac{n-1}{2}\right)}{\Gamma\left(\frac{n_A-1}{2}\right)\Gamma\left(\frac{n_B-1}{2}\right)\Gamma\left(\frac{1}{2}\right)} \int_D (1-x-y)^{\frac{n_A-1}{2}-1} x^{\frac{n_B-1}{2}-1} y^{\frac{1}{2}-1} dxdy, \tag{24}$$

where

$$D = \{(x,y) \in [0,1]^2 \colon x+y \leq 1, \lambda_A^r(1-x-y) + \lambda_B^r x + \lambda_{\text{neg}}^r y \leq 0\}.$$

The integral can thus be written as

$$\int_0^1 \int_0^{u(y)} (1-x-y)^{\frac{n_A-3}{2}} x^{\frac{n_B-3}{2}} y^{-\frac{1}{2}} dxdy,$$

where $u(y) = \max\left(\min\left(1-y, -\frac{\lambda_A^r + y(\lambda_{\text{neg}}^r - \lambda_A^r)}{\lambda_B^r - \lambda_A^r}\right), 0\right)$, in the case where $\lambda_B^r - \lambda_A^r > 0$ (if $\lambda_B^r - \lambda_A^r < 0$, change the roles of A and B). If $\lambda_A^r = \lambda_B^r$, then

$$\mathbb{P}_0(Z_{r,1} \leq 0) = \frac{\Gamma\left(\frac{n-1}{2}\right)}{\Gamma\left(\frac{n-2}{2}\right)\Gamma\left(\frac{1}{2}\right)} \int_0^u (1-x)^{\frac{n-4}{2}} x^{-\frac{1}{2}} dx, \tag{25}$$

where $u = \max\left(\min\left(1, \frac{\lambda_A^r}{\lambda_A^r - \lambda_{\text{neg}}^r}\right), 0\right)$, in the case where $\lambda_{\text{neg}}^r - \lambda_A^r > 0$ (otherwise switch the roles of λ_{neg}^r and λ_A^r).

For $n \approx 350$ and larger, the computation of the integrals in (**??**) and (**??**) become numerically unstable. However, this does not seem to be problematic since for such large values of n, the distribution of R under H_0 is close to a point mass at 1 (cf. Figure **??**).

./Figures/10_13_F_R.png ./Figures/160_170_F_R.png

Figure 1: $F_R^0(r)$ for $(n_A, n_B) = (10, 13)$ and $(n_A, n_B) = (160, 170)$.

Figure **??** shows how the distributions change when n_A, n_B vary while n is fixed. It seems that low quantiles are not much affected by unbalancedness of (n_A, n_B).

./Figures/50_50_F_R.png ./Figures/95_5_F_R.png

Figure 2: $F_R^0(r)$ for $(n_A, n_B) = (50, 50)$ and $(n_A, n_B) = (95, 5)$.

4.3 Likelihood ratio test idea – not promising

As above, let $H_0 : \mu = (c, \ldots, c)^T$ for some $c \in \mathbb{R}$ (no signal) and define the alternative hypothesis $H_1 : \mu_A = (c_1, \ldots, c_1)^T, \mu_B = (c_2, \ldots, c_2)^T$ for some $c_1, c_2 \in \mathbb{R}$ with $c_1 \neq c_2$ (signal). These hypotheses can be equivalently described by setting $\xi_A = \xi_B = 0$ in (**??**) and

$$H_0 : \xi_{\text{neg}} = 0, \quad H_1 : \xi_{\text{neg}} > 0.$$

Denoting by $z \in \mathbb{R}$ a realisation of Z, we get a generalised likelihood-ratio of the form

$$\frac{\sup_{\xi_{\text{neg}}>0} \sum_{j=0}^{\infty} K_j(\xi_{\text{neg}}) \frac{1}{2^{(n-1+2j)/2}\Gamma((n-1+2j)/2)} \left(\frac{z}{\sigma^2}\right)^{(n-1+2j)/2-1} \exp\left(-\frac{z}{2\sigma^2}\right)}{\sum_{j=0}^{\infty} K_j(0) \frac{1}{2^{(n-1+2j)/2}\Gamma((n-1+2j)/2)} \left(\frac{z}{\sigma^2}\right)^{(n-1+2j)/2-1} \exp\left(-\frac{z}{2\sigma^2}\right)}$$

following [On the Distribution of Linear Combinations of Non-central Chi-Squares, By DAVID A. HARVILLE]. Here, K_j are functions depending on n_A, n_B and ξ_{neg}, but not on z.

This quantity however is not independent of σ^2, which is why the likelihood ratio approach seems to be a bad ansatz if we want to avoid estimating σ^2.

5 Hypothesis Testing

Following the arguments in the previous section, we argue the following. In the ratio test, we look at relative reduction in msep instead of absolute difference. The criterion will instead be to continue as long as the relative reduction $r^* \leq r^0_\alpha$. Since finding the quantile can be difficult for general distributions of of msep. We can make it simpler in our case by looking at a linear combination of chi-squares. The general CDF can be computed from [**?**] and computing the CDF for R is equivalent to

$$F^0_R(r) = \mathbb{P}(R^0 \leq r) = \mathbb{P}(Z^0_r \leq 0)$$

In practice, one should compute r^* and then $\mathbb{P}(Z^0_{r^*} \leq 0)$.

6 Limiting distribution of Z

In the limit $n_a, n_b \to \infty$, we get that the eigenvalues of all tend to 0 except for $\lambda_n eg = -1$ in this limiting case, the distribution for Z_∞ is

$$Z_\infty \stackrel{d}{=} -\sigma^2 \bar{\chi}^2_1$$

Can we say something about this? Check that this is actually true.

Limit update (I think we were wrong about this the last time...) Let $n \to \infty$ and assume that $n_A/n \to a \in (0,1)$, $n_B/n \to 1-a$. We want to investigate the weak limit of $Z_{r,1}$ under H_0, i.e. the limit of

$$\tilde{\lambda}^r_A \frac{\chi_{n_A-1}}{n_A-1} + \tilde{\lambda}^r_B \frac{\tilde{\chi}_{n_B-1}}{n_B-1} + \lambda^r_{\text{neg}} \bar{\chi}_1, \tag{26}$$

where all chi-squares are independent and

$$\tilde{\lambda}_A^r = \frac{(n_A+1)(n-1) - r(n+1)(n_A-1)}{n-1},$$
$$\tilde{\lambda}_B^r = \frac{(n_B+1)(n-1) - r(n+1)(n_B-1)}{n-1}, \quad \lambda_{\text{neg}}^r = -r\frac{n+1}{n-1}.$$

The first two standardised chi-squares in (**??**) each converge to 1 in probability. Moreover,

$$\lambda_{\text{neg}}^r \to -r,$$
$$\tilde{\lambda}_A^r = \frac{2n_B}{n-1} + \frac{(1-r)(n+1)(n_A-1)}{n-1} \to \begin{cases} 2(1-a), & r=1 \\ -\infty, & r>1 \\ +\infty, & r<1 \end{cases}$$

and similarly

$$\tilde{\lambda}_B^r \to \begin{cases} 2a, & r=1 \\ -\infty, & r>1 \\ +\infty, & r<1. \end{cases}$$

Hence, under H_0, we get convergence in distribution by the Slutzky Theorem

$$Z_{r,1} \to Z_{r,1}^\infty = \begin{cases} 2-\bar{\chi}_1, & r=1 \\ -\infty, & r>1 \\ +\infty, & r<1. \end{cases}$$

Since $Z_{r,1}^\infty$ has no mass at 0 for any $r>0$, this implies that

$$F_R^0(r) = \mathbb{P}_0(Z_{r,1} \le 0) \to \mathbb{P}_0(Z_{r,1}^\infty \le 0) = \begin{cases} \mathbb{P}(\bar{\chi}_1 \ge 2), & r=1 \\ 1, & r>1 \\ 0, & r<1. \end{cases}$$

This shows that the ratio R converges in probability to 1 under H_0. This is in line with the numerical results in Figure **??**. Note that the limit does not depend on a.

7 Building Trees

7.1 Breiman minimal cost-complexity pruning

Breiman's method of minimal cost-complexity builds a full tree and sequencially prunes nodes to construct a tree that is less complex and that is supposed to generalize better.

If we denote T as a tree and define $R(T)$ as the weighted loss contributions from the leaves of T, we can define a penalized loss $R_\alpha(T) = R(T) + \alpha|T|$, where $|T|$ is the number of leaves of the tree T and α is some hyper-parameter that penalizes complexity. Breiman's algorithm produces the best **subtree** for a given α that minimizes $R_\alpha(T)$. Furthermore, it can be shown that the best trees are nested subtrees T_k of T such that each is optimal for an interval of α. So there is a sequence $\{\alpha_k\}$

$$0 = \alpha_0 < \alpha_1 < \cdots < \alpha_m < \infty \tag{27}$$

such that T_i is the optimal tree w.r.t $R_\alpha(\cdot)$ for $\alpha \in [\alpha_i, \alpha_{i+1})$. Here it should be clear that $T_0 = T$ and T_m is the trivial tree. What this essentially means is that, given some value $\alpha \in [\alpha_i, \alpha_{i+1})$ the tree that minmizes $R_\alpha(\cdot)$ is T_i and T_{i+1} is a subtree of T_i, so starting at $\alpha = 0$ and increasing α will produce a sequence of trees that are pruned subtrees of the preceding until the trivial tree T_m is reached.

The procedure for attaining $\{\alpha_k\}$ and T_k is as follows. Given a non-trivial tree T_i, denote all non-terminal nodes with some t and calculate the gain $g(t, T^{(t)}) = \frac{R(t) - R(T^{(t)})}{|T^{(t)}| - 1}$ for all t, where $R(t)$ is the loss at the node t treated as a trivial tree and $R(T^{(t)})$ is the loss of the tree that is rooted at t. Then $\alpha_i = \min_t g(t, T^{(t)})$. Pruning all nodes where the $g(\cdot) = \alpha_i$ produces a new tree T_{i+1}. Now perform the same procedure on T_{i+1} and so on until the trivial tree is reached. This results in a sequence $\{\alpha_k\}$ and a sequnce of nested trees $\{T_k\}$. Two remarks can be made. Firstly, the procedure handles ties in gain by choosing the smallest tree as next in line. Second, the gain is attained from simple algebra by solving for α in the equation $R_\alpha(t) - R_\alpha(T^{(t)}) = 0$.

7.2 Adaptive Boosting

When boosting regression trees, there are in general two parameters that need to be decided. First, the complexity of the weak learners, e.g. number of leaves ofr regression trees, at each boosting step and second the stopping criterion, that is, step k to stop boosting. The complexity of the tree is choosen heuristically. However, the stopping criterion is typically decided using cross validation. Where the training is stopped as soon as the out-of-sample error starts increasing. Adaptive Boosting Trees(ABT) tackles this problem by proposing an algorithm for boosting with a built-in stopping rule that is based on the complexity of the weak learner. That is, it is sufficient to choose a parameter J and the procedure will produce boosted trees with at most complexity J and will stop.

ABT leverages Breiman's pruning method (see **??**) to produce learners of at most J leaves. That is, at each boosting iteration a tree is fitted by choosing the subtree T_k of T such that

$$|T_{k+1}| > J \text{ and } |T_k| \leq J.$$

If $|T_k| = 1$ then the procedure stops. The reason for stopping when this criterion is met, is that the intercept will be identically zero, thus, boosting after this criterion is met will further produce intercepts of zero which is unecessary. Note that in the ABT paper they present a general framework for choosing the response that should be fit at each iteration given a deviance loss. However when an L^2 loss is considered, the reponse at each iteration is the residuals.

A problem raised in the ABT paper is that their procedure sometimes stops early, "stuck in local optima". They propose using a bagging fraction β to overcome this issue.

Can we say that the algorithm is guaranteed to stop if α is guaranteed to stop in finite time? That is $\tau = \inf k|\alpha_k = 0 < \infty$ in probability or almost surely? If I haven't made a mistake it seem like it does fluctutate, i.e not monotonic but it does decrease rather fast to very samll numbers.

7.2.1 Varying sample sizes

It is interesting to to understand how this method performs for ranges of different sample sizes. We simulate data for studying this in the following way. First we define the underlying mean as

$$\mu(x) = 10 + 10\mathbf{1}_{\{x>0.7\}} - 25\mathbf{1}_{\{x<0.2\}} + 3\mathbf{1}_{\{x<0.5\}}.$$

This gives us 4 levels for the response.The data point is then distributed in the following way $X \sim Unif([0,1])$ and $Y|X = x \sim N(\mu(x), \sigma^1)$, for some σ that controls the level of noise. For the different sample sizes we get the following statistics.

TODO:

1. Plot of MSEP for different sample sizes
2. Plot of MSEP for number of iterations for these sample sizes

7.2.2 Connection to MSEP p-values

Is there a connection between the p-values we calculate and the stopping mechanism in ABT? A simulation study gives the following results.

7.3 Gradient boosting with MSEP inference

We propose a method for stopping tree building, using inference. Where we decide on a significance level $\tilde{\alpha}$ and bound the tree building process using Bonferroni type arguments. The stopping rule is then, to stop when the cummulative sum of the p-values at each split exceeds $\tilde{\alpha}$. However, one caveat is that we need to decide on a splitting sequence for which we sum the p-values. Heuristically, we can choose the trees produced by Breiman's algorithm as a sequence splits.

An alternative method would be to boost residuals using depth 1 trees as weak learners. This solves the problem of choosing a sequence, since the sequence is the natural boosting iterations. Then, a stopping criterion is to stop boosting when the cummulative sum of p-values at each boosting iteration exceeds $\tilde{\alpha}$. It is then possible to resuse our hypothesis test. However, we have not thought out how to incorporate more complex learners. Maybe Breiman + inference + boosting. In some sense altering ABT?

An intial simulation study results in the following figures.